

Low frequency ship vibration isolation using the band gap concept of sandwich plate-type elastic metastructures

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ABSTRACT

Elastic metamaterial is a type of artificial composite material/structure with unique subwavelength physics property, which shows attractive potential application in low-frequency sound and vibration control. In this paper, a ship vibration isolation method using the band gap concept of sandwich plate-type elastic metastructures is proposed to solve the low frequency vibration and noise control of marine power system. The flexural wave propagation and low-frequency vibration isolation characteristics of sandwich plate-type elastic metamaterials are investigated by using the finite element method combination with Solid-Shell multi-physics coupling model and Bloch theory. Numerical results demonstrate that the proposed sandwich plate-type elastic metastructures possess significant low-frequency subwavelength flexural wave band-gap and vibration isolation characteristics under the premise of structural strength and stiffness. The formation mechanism of flexural band gap is mainly attributed to the mode coupling of the local resonant mode and the Lamb wave mode of sandwich plate. The experimental measurements of flexural vibration transmission spectra were conducted to validate the accuracy and reliability of the numerical calculation method. In addition, the influences of geometrical parameters on the flexural wave band gap are studied. On this basis, the potential application of the proposed ship vibration isolation method in the marine power system is explored.

1. Introduction

With the rapid development of marine traffic industry, ship vibration and noise pollution is becoming more and more serious, high efficient vibration and noise control technologies such as passive control, active control, semi-active control and hybrid control technology (Kandasamy et al. (2016); Lin et al. (2009); Zhang et al. (2017)) are crucially important. Ship vibration and noise radiation is mainly caused by various marine power system and equipment excitation, propeller excitation and wave excitation. During the operation of marine power system and equipment, mechanical parts will generate large power and broadband excitation sources. These excitation sources will transmit vibration in the form of elastic waves through hull plate to cause structural vibration and noise radiation. In addition, in the vibration transmission process of hull plate, elastic waves are mainly manifested as flexural wave, longitudinal wave and torsion wave. The longitudinal wave and torsion wave are both high impedance waves with high wave number and they can hardly interact with sound waves, while the

flexural wave is low impedance wave and can interact with sound waves. Therefore, the suppression of flexural wave propagation is of great significance to ship vibration and noise control.

Elastic metamaterial is a type of artificial composite material/structure with unique physics property such as negative mass density (Liu et al., 2000; LIU et al., 2005; Sheng et al., 2007), negative modulus (Fang et al., 2006; Lee et al., 2010) and subwavelength band gap characteristic (Cai et al., 2021; Peng and Frank Pai, 2014; Yuan et al., 2021), which shows attractive potential application in low-frequency sound and vibration control. The flexural vibration of thin plate is the main factor that causes the structure noise radiation. Therefore, a large number of studies have been carried out to investigate the flexural wave propagation and vibration isolation characteristics in plate-type elastic metamaterial (Bilal and Hussein, 2013; Jung et al., 2020; Langfeldt and Gleine, 2019; Xiao et al., 2012). Pennec et al. (2008) and Wu et al. (2008) respectively proposed a plate-type elastic metamaterial structure formed by a finite height cylinder periodically distributed on an isotropic thin plate. The plate-type elastic metamaterial structure can

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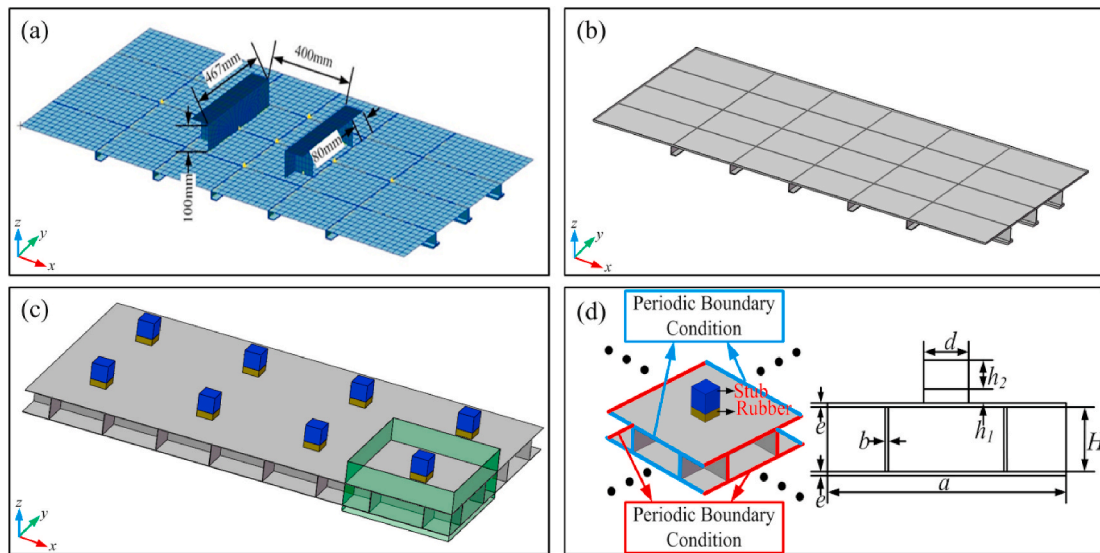


Fig. 1. (a) The representative marine power system; (b) Schematic diagram of the hull grillage; (c) Schematic diagram of sandwich plate-type elastic metastructure; (d) Unit cell of sandwich plate-type elastic metastructure.

Table 1
The geometrical parameter of the hull grillage.

Geometric parameters	Dimension/mm
Length of tank top	2000
Width of tank top	1000
Thickness of tank top	6
Longitudinal (T shape)	TN 50 × 50 × 5 × 7
Reverse frame (L shape)	L 30 × 3

generate a low-frequency band gap. Badreddine Assouar and Oudich (2012) demonstrated that the band gap width of the plate-type acoustic metamaterial with a double-side stub is about twice that of the classical one-side stubbed plates. Li et al. (2015); Li et al. (2017) studied the low frequency and broadband elastic wave attenuation and vibration

suppression of the plate elastic metamaterial. Miranda et al. (2020) studied the complex band structure of metamaterial plate with attached spring-mass resonators by using the extended plane-wave methods.

Above plate-type elastic metamaterials are generally constituted of locally resonators deposited on homogenous monolayer plate. Recently, the wave propagation characteristics and vibration behaviors of sandwich plate-type elastic metamaterials have been paid increasing attention. Wu et al. (2015) carried out theoretical modeling for corrugated

Table 2
Dimensions of proposed sandwich plate-type metastructures (mm).

Geometric parameter	a	b	d	e	H	h_1	h_2
Dimension	200	0.9	50	0.9	40	20	50

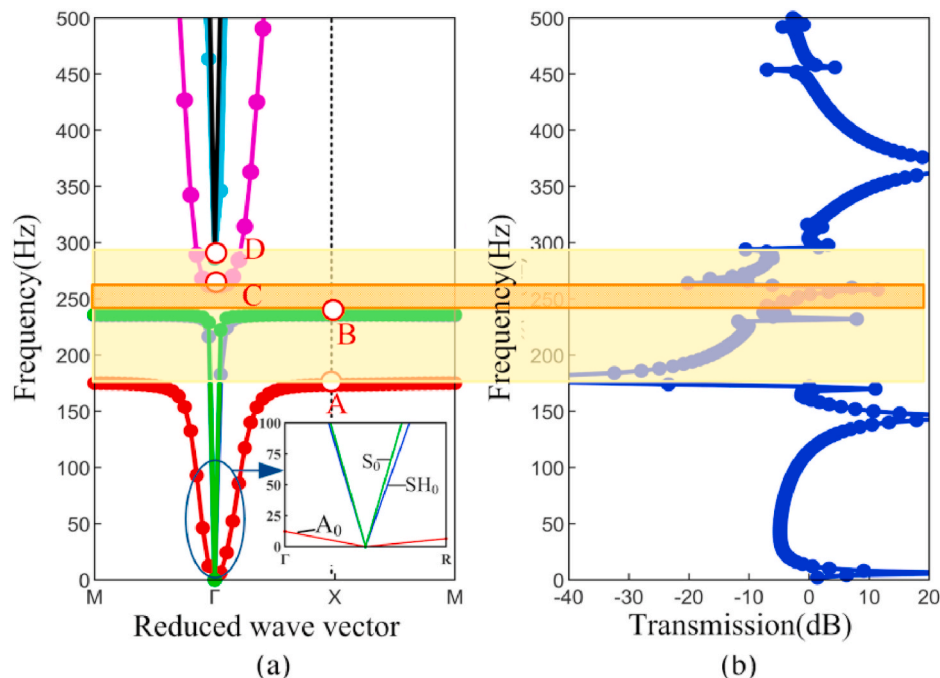


Fig. 2. Band structure (a) and frequency response characteristic (b) of the proposed sandwich plate-type elastic metastructure.

Table 3
Material parameters of proposed sandwich plate-type metastructures.

Material	Young's modulus (Pa)	Poisson's ratio	Density (kg/m ³)
Rubber	2×10^7	0.47	1300
Steel	2×10^{11}	0.3	7800

sandwich plates and calculated band gap characteristics by using the spectral element method. They compared the theoretical calculation results with the finite element calculation results to confirm the correctness of spectral element method. Song et al. (2019) respectively calculated the band structure and transmission characteristics of sandwich metamaterial plate and the bare sandwich plate to study the effect of periodic design on vibration and sound radiation. Wang et al. (2021) studied the band structure and band gap formation mechanism of the metamaterial sandwich plate with plate-type resonator by using numerical and experimental methods.

In addition to the elastic wave propagation characteristic and band gap formation mechanism in elastic metamaterials, increasing attention has been paid to explore the engineering application of elastic metamaterials in the vibration and noise reduction of ship and offshore structures. Li et al. (2019) studied the flexural wave band gaps and vibration attenuation characteristics of periodic bi-directionally orthogonal stiffened plates. The flexural wave band gap of periodic bi-directionally orthogonal stiffened plates is mainly concentrated in medium and high frequency, which is difficult to solve the low frequency ship vibration isolation. Ruan et al. (2021) designed and analyzed a single-phase spiral-shaped local resonant phononic crystal for the sake of vertical vibration excitation isolation of a certain type of ship. This work is an active exploration of phononic crystals into the field of ship vibration and noise control. However, with the introduction of spiral-shaped phononic crystal, the mechanical properties such as strength and stiffness of ship structures are difficult to guarantee.

In the present work, a ship vibration isolation method using the band gap concept of sandwich plate-type elastic metastructures is proposed to solve the low frequency vibration and noise control of marine power system and promote the engineering application of elastic metamaterials in the field of ship and offshore structures. The flexural wave propagation and low-frequency vibration isolation characteristics of sandwich plate-type elastic metastructure consisting of a sandwich plate with periodic stepped resonators are numerically investigated. The experimental measurements of the acceleration frequency responses were conducted to validate the accuracy and reliability of the numerical calculation method. In addition, the influences of geometrical parameters on the flexural wave band gap are studied. Finally, the potential

application of the proposed sandwich plate-type elastic metastructure in the vibration control of marine power system is explored.

2. Physics model and calculation method

2.1. Sandwich plate-type elastic metastructures

In this paper, the representative marine power system is considered and installed on the foundation and hull grillage as shown in Fig. 1(a). The hull grillage is composed of the tank top and the longitudinal frames as well as the reverse frames as shown in Fig. 1(b). The geometrical parameter of the hull grillage is illustrated in Table 1. The longitudinal frame space is 250 mm and the transverse frame space is 300 mm. According to the principle of similarity theory, the equivalent design of grid sandwich plate structure is carried out combination with the parameter equivalence such as the neutral axis height, section area and section modulus of the scale-down hull grillage. The bending strength

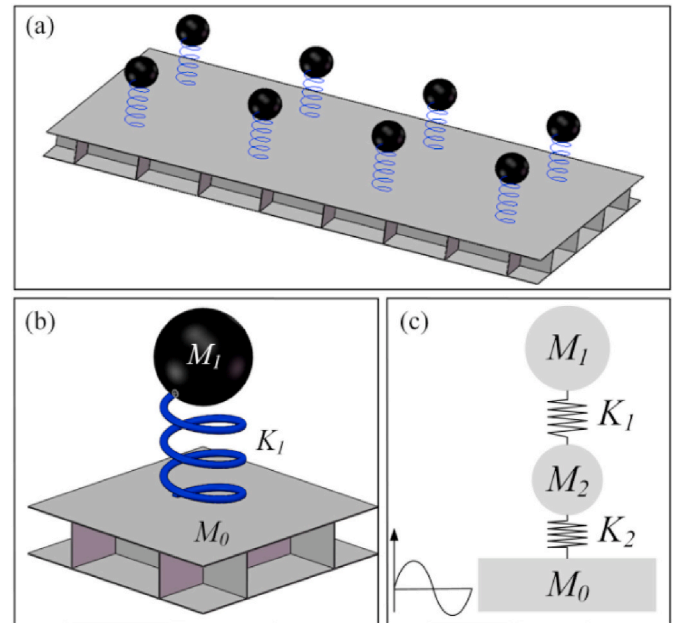


Fig. 4. (a) Schematic diagram of local resonant sandwich plate; (b) The unit cell of the spring mass system; (c) Two degrees of freedom 'spring-mass' model of the plate-type elastic metamaterial.

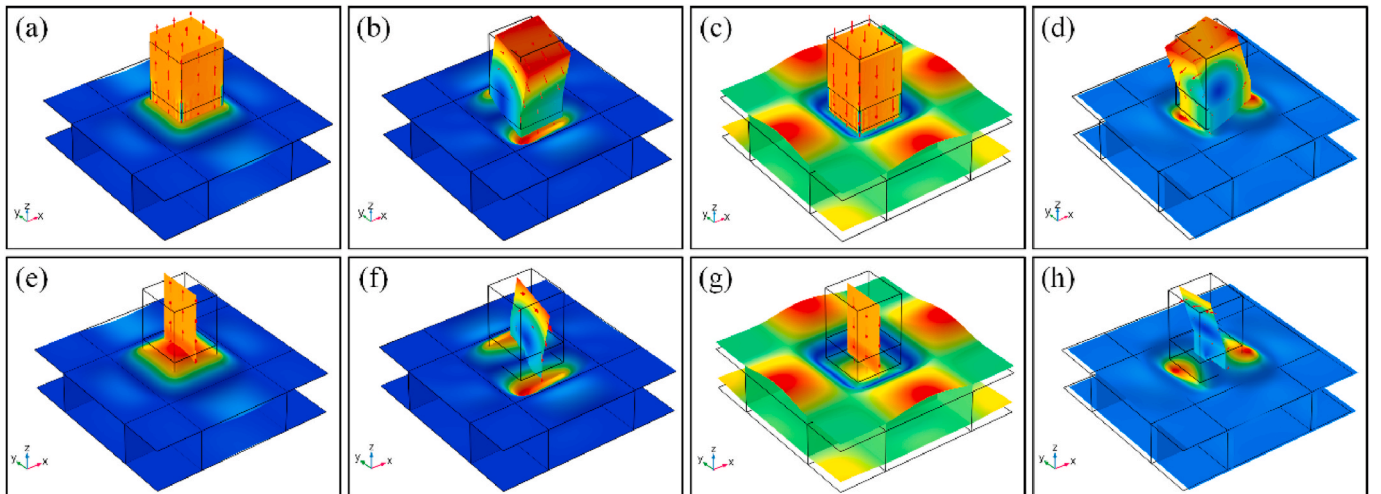


Fig. 3. Eigenmodes and corresponding cross sections of unit cell marked as A, B, C, D in Fig. 2(a).

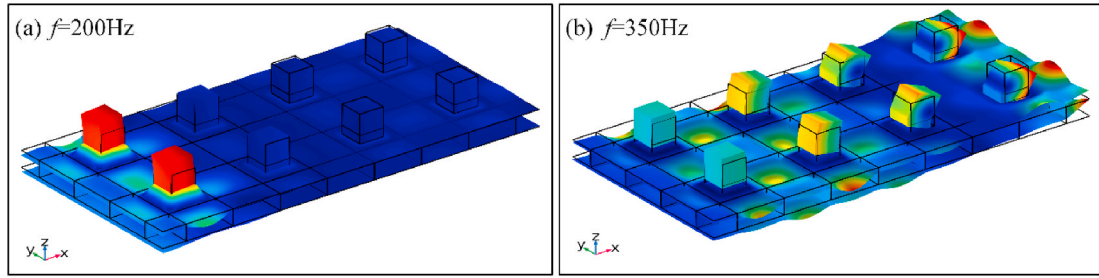


Fig. 5. Flexural vibration transmission for different frequencies (a) $f = 200$ Hz; (b) $f = 350$ Hz.

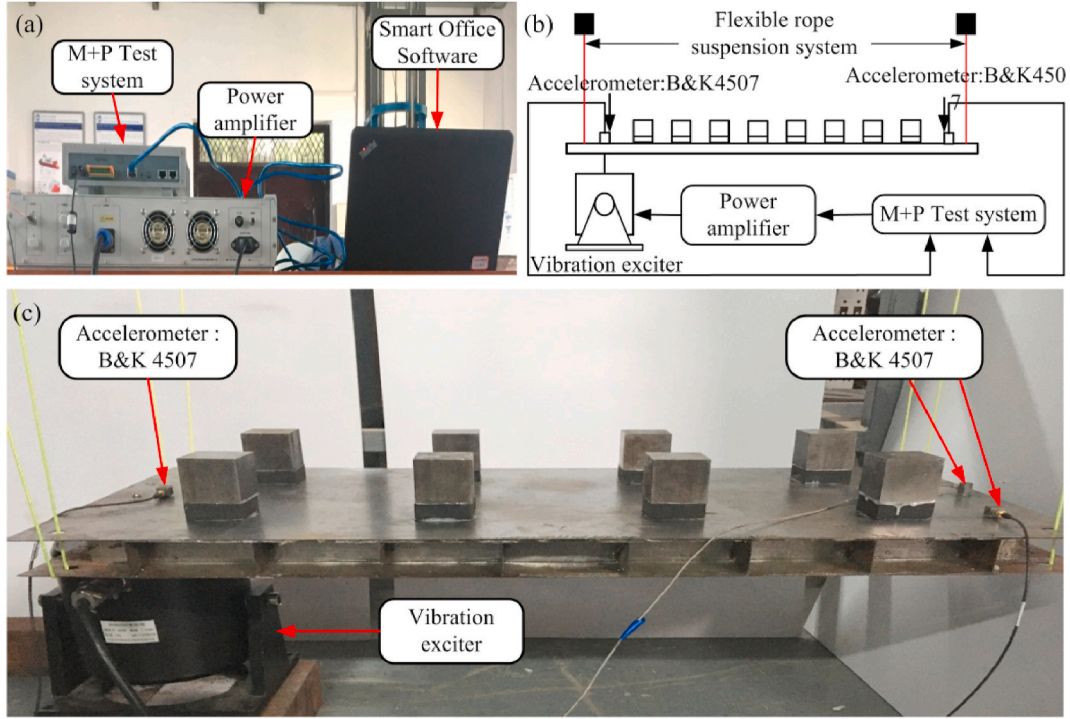


Fig. 6. (a) The experimental measurement system; (b) The experimental measurement principle; (c) The fabricated sample of the sandwich plate-type metastructure.

and stiffness of the equivalent sandwich plate is higher than those of the hull grillage, which demonstrates that the equivalent sandwich plate can guarantee the structural mechanical property requirements. In order to simultaneously realize the low-frequency vibration isolation performance of marine power system, a sandwich plate-type elastic metastructure is designed based on the locally resonant metamaterial theory. The sandwich plate-type elastic metastructure considered here is constituted of the equivalent sandwich plate and periodically attached local resonators along x -direction and y -direction as shown in Fig. 1(c). The z axis is perpendicular to the sandwich plate and parallel to the periodic stubs axis. Fig. 1(b) shows the schematic diagram of unit cell of the proposed sandwich plate-type elastic metastructure. In the unit cell, the geometrical parameters of the considered metastructures are defined as follows: the lattice constant of the sandwich plate-type metastructures respectively are defined as a , the plate thickness, the web thickness and height are denoted by e , b and H respectively, the side length of the rubber block and the steel stub is d , and the height of the rubber block and the steel stub are h_1 and h_2 respectively.

2.2. Calculation method

In order to investigate the flexural wave propagation and vibration isolation characteristics of the proposed sandwich plate-type elastic

metastructures, the dispersion relations and the transmission power spectra of sandwich plate-type elastic metastructures are calculated by using an efficient finite element method combination with Solid-Shell multi-physics coupling model and Bloch theory via the COMSOL Multiphysics 5.4. The local stepped resonators are built as solid structures by using the Lagrange-Quadratic element type in the Solid Stress-Strain Application Mode. For the elastic wave propagation in solid structures, the wave equations can be written as:

$$\sum_{j=1}^3 \left\{ \frac{\partial}{\partial x_i} \left(\lambda \frac{\partial u_j}{\partial x_j} \right) + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right] \right\} = \rho \frac{\partial^2 u_i}{\partial t^2} \quad i, j = x, y, z \quad (1)$$

where ρ is the mass density; u and v denote the displacement vector; t is the time, λ and μ are the Lamé constants, x , y and z represent the coordinate variables in Cartesian coordinates system respectively.

Besides, the equivalent grid sandwich plate is established as shell structures in the Shell Application Mode. The thicknesses as well as the material properties of the shell structures should be further defined. The shell element type is defined as the MITC type, which can be used for analyzing both thin (Kirchhoff theory) and thick (Mindlin theory) shells. This MITC type shell theory is developed from the Mindlin-Reissner shell theory. The Mindlin-Reissner shell theory improves the assumption of the straight normal line in the Kirchhoff shell theory. Taking into ac-

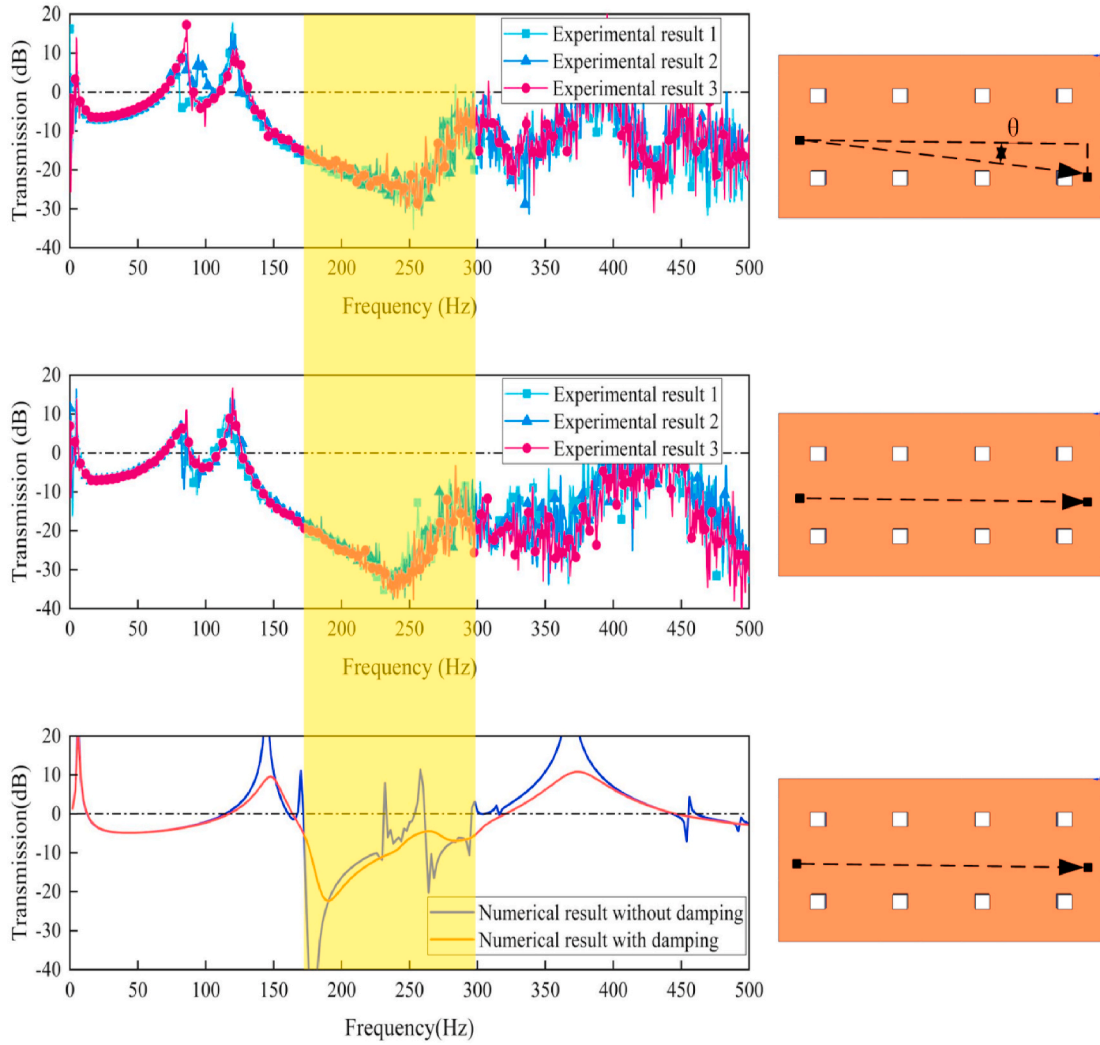


Fig. 7. Experimental validation of flexural vibration frequency responses.

count the transverse shear deformation and moment of inertia of the plate, the flexural wave equation of the plate in the Mindlin-Reissner shell theory can be written as follows:

$$-\rho h \frac{\partial^2 w}{\partial t^2} = \frac{\partial^2}{\partial x^2} \left[D \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} \right) \right] + 2 \frac{\partial^2}{\partial x \partial y} \left[D \frac{\partial^2 w}{\partial x \partial y} \right] - \frac{\partial^2}{\partial t^2} \left(\frac{\partial}{\partial x} \left(\left(\frac{\kappa \rho}{G} + \rho J \right) \frac{\partial w}{\partial x} \right) + \frac{\partial}{\partial y} \left(\left(\frac{\kappa \rho}{G} + \rho J \right) \frac{\partial w}{\partial y} \right) \right) + \left(\frac{\kappa \rho}{G} J \right) \frac{\partial^4 w}{\partial t^4} \quad (2)$$

where w is the transverse deflection; h is the thickness of the plate; D denotes flexural rigidity; G is the modulus of rigidity; κ is the shear correction factors; and J represents moment of inertia.

$$G = \frac{E}{[2(1+\nu)]}, \kappa = 20 \frac{(1-\nu)}{(24+25\nu+\nu^2)}, J = \frac{h^3}{12} \quad (3)$$

In an attempt to calculate the dispersion relations and displacement fields of the eigenmodes, only one representative unit cell of the proposed sandwich plate-type elastic metastructures need to be considered due to spatial symmetry and translational periodicity according to energy-band theory and solid lattice theory. Stress-free boundary conditions are defined for free surfaces in the z -direction, while periodic boundary conditions are applied at the boundaries with the Bloch theorem as follows:

$$u_i(x+a, y+a) = u_i(x, y) e^{i(k_x a + k_y a)} \quad i = x, y \quad (4)$$

where x and y are the coordinate components of position vector; k_x, k_y are the coordinate components of the Bloch wave vector in the first Brillouin zone. The band structures and the eigenmodes can be obtained by calculating the eigenvalues and eigenvectors of the wavevectors. In this analysis, the displacement is assumed to be a time harmonic of angular frequency ω . According to the finite element principle, the elastic wave dynamic formulation of the typical element can be transformed into the standard generalized eigenvalue formulation with discrete significance:

$$(K - \omega^2 M)u = 0 \quad (5)$$

where K, M and u represent the stiffness matrix, mass matrix and eigenvector matrix, respectively.

To calculate the transmission spectra of sandwich plate-type elastic metastructures, the finite array system formed by 4 unit cells along the x -direction and 2 unit cells along the y -direction is considered as shown in Fig. 2(c). The sine sweep acceleration excitation resource a_{in} is imposed on one end of the sandwich plates. The flexural vibration direction of the excitation resource is along the z -direction and the flexural wave propagates along the y -direction. The transmitted acceleration response a_{out} is detected on the other end of the sandwich plates. The transmission spectra T is defined as

$$T = 20 \log \frac{|a_{out}|}{|a_{in}|} \quad (6)$$

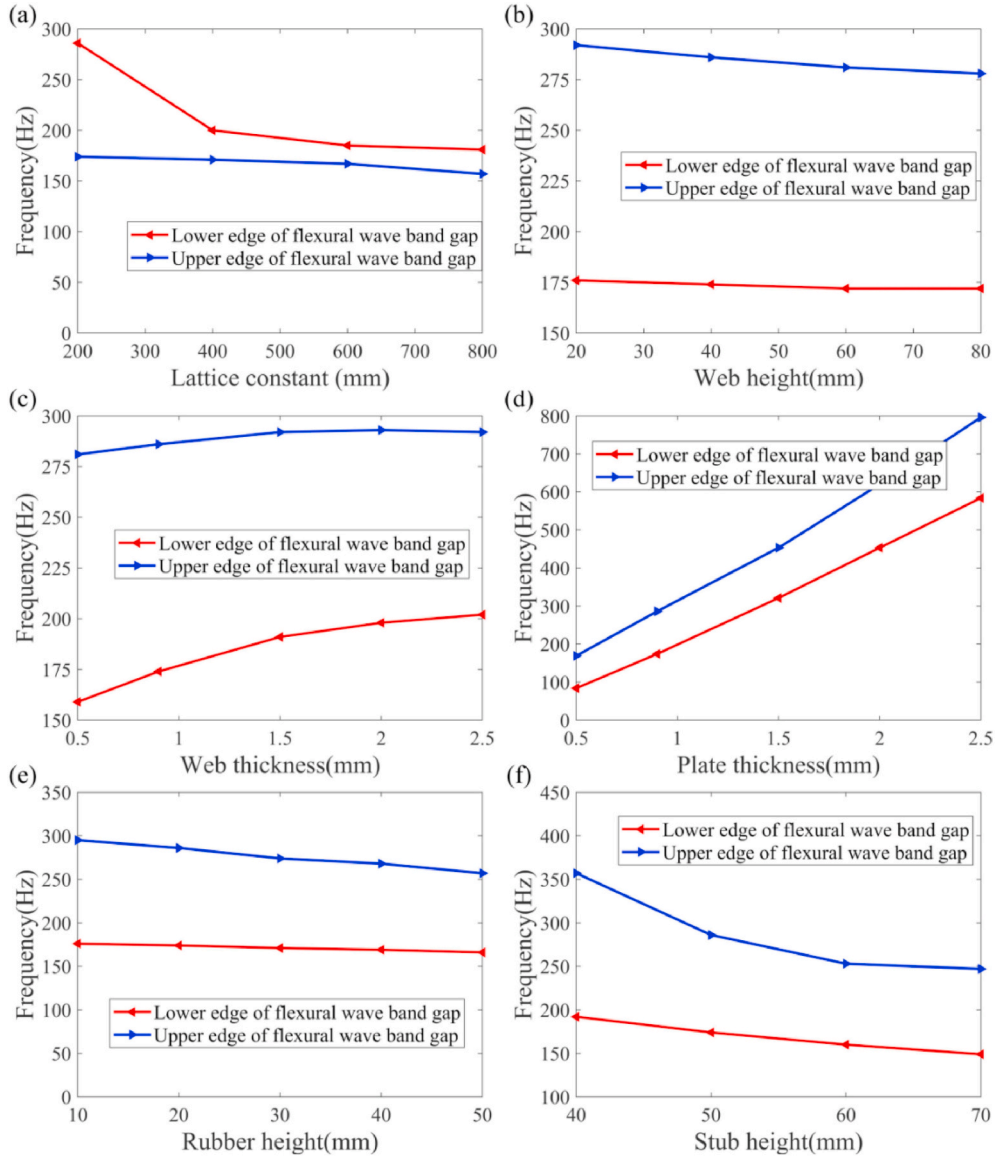


Fig. 8. Effects of geometrical parameters on the band gap: (a) lattice constant; (b) web height; (c) web thickness; (d) plate thickness; (e) rubber height; (f) stub height.

Table 4
Dimensions of sandwich plate-type elastic metastructures at the real ship scale (mm).

Geometric parameter	a	b	d	e	H	h_1	h_2
Dimension	400	1.8	100	1.8	80	40	100

3. Results and discussion

3.1. Dispersion relation and transmission spectra

To study the flexural wave propagation and band gap characteristics of the proposed sandwich plate-type elastic metastructures, the dispersion relation is obtained as shown in Fig. 2(a). The horizontal axis represents the reduced wave vector along the boundary of the irreducible Brillouin zone and the vertical axis is the wave frequency. In the calculation, the equivalent sandwich plate and the steel stubs are defined as steel, while the rubber blocks are defined as rubber. Geometric parameters and material parameters are shown in Tables 2 and 3

respectively. It can be found from Fig. 2(a) that there are six bands in the frequency range from 0 to 500 Hz, where one complete low-frequency band gap (236–262 Hz in orange region) and a flexural wave low-frequency band gap (174–286 Hz in canary region) are involved. On the basis of plate wave theory, a finite thickness plate admits the propagation of a number of anti-symmetric (A mode) and symmetric (S mode) Lamb waves and shear-horizontal (SH mode) waves with cutoff frequencies, including three fundamental plate-modes (i.e., A_0 , S_0 , and SH_0 modes) and infinite overtones (A_n , S_n , and SH_n modes, $n = 1, 2, 3, \dots$). In the band structure, the dispersion relation branches originating from Γ point can be identified as the A_0 , S_0 , and SH_0 (i.e., the lowest anti-symmetric, symmetric, and shear-horizontal) modes in the plate. In addition, we can observe a large number of flat modes in the band structures due to the locally resonant effect. The location of sub-wavelength flexural band gap is two orders of magnitude lower than that of Bragg scattering band gap.

To further demonstrate the existence of the low-frequency band gap in the proposed sandwich plate-type elastic metastructures, the flexural vibration frequency response of the 4×2 unit cells periodic sandwich plate-type metastructures is calculated as shown in Fig. 2(b). During the

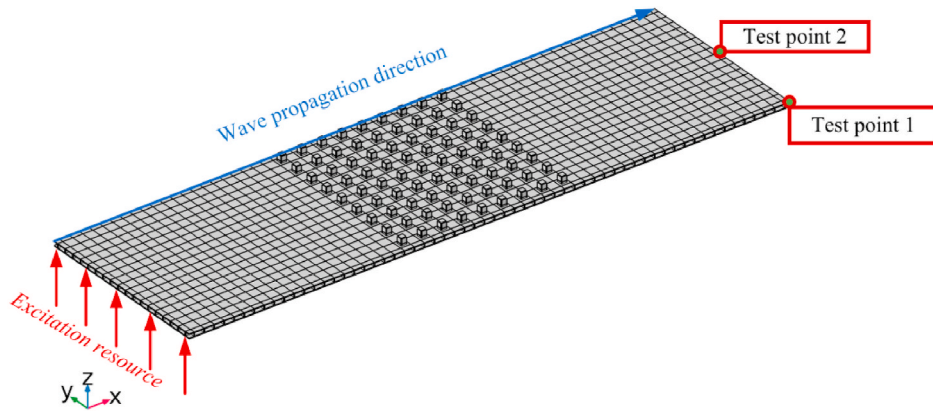


Fig. 9. The schematic diagram of full-scale sandwich plate-type metastructures in the vibration control application of marine power system.

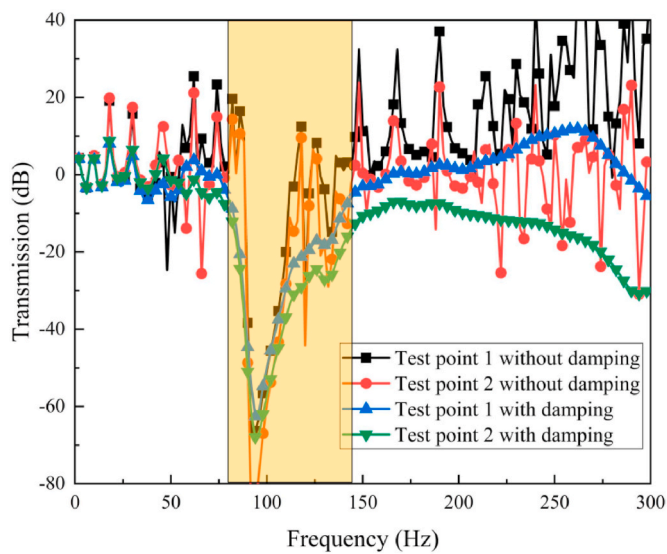


Fig. 10. Flexural vibration frequency responses of full-scale sandwich metastructures.

calculation, the geometrical and material parameters remain unchanged. It can be observed that there is a significant broadband flexural vibration attenuation zone in the frequency response of bending vibration. The location and width of the flexural vibration attenuation exhibit very good agreement between the dispersion relation and the flexural vibration frequency response, which verifies the correction and feasibility of Solid-Shell multi-physics coupling method. The results demonstrate that the proposed sandwich plate-type metastructure have flexural wave band gap in the subwavelength frequency range, which can inhibit low frequency flexural wave propagation and achieve the flexural vibration isolation.

In an attempt to further reveal the formation mechanisms of the flexural wave band gap in the proposed sandwich plate-type metastructures, the displacement fields of the eigenmodes as well as their cross sections cutting by y - z planes in the specific wave vectors are calculated as shown in Fig. 3. Fig. 3(a) and (d) respectively correspond with the starting and cutoff frequencies of the flexural wave band gap. It can be noted that eigenmodes A as shown in Fig. 3(a) is mainly represented by the coupling locally resonant mode between the Lamb wave of sandwich plates and the longitudinal vibration mode of local resonators, which leads to the emergence of the locally resonant flexural wave band gap. We can find from Fig. 3(d) that the eigenmodes D mainly shows the coupling mode in which the local resonator twists about the y -direction

and the sandwich plate moves transversely along the x -direction. Fig. 3 (b) and (c) correspond with the starting and cutoff frequencies of the complete band gap respectively. Mode B is similar to mode D, but the stepped resonator of the mode B twists about the X -axis. Therefore, mode D has a certain inhibitory effect on the propagation of the flexural wave along the Y -axis, while mode B is conducive to the propagation of the flexural wave along the Y -axis. The displacement field of the eigenmodes C as shown in Fig. 3(c) is mainly attributed to the local resonator and the sandwich plate moving towards each other in the z direction. From the above analysis, we infer that the generation of the low-frequency band gaps in the proposed sandwich plate-type metastructures is mainly due to the coupling between the local resonant modes and the asymmetric Lamb wave modes propagating in all directions.

The vibration energy is concentrated near the local resonator, and the locally resonant sandwich plate-type elastic metamaterial can be equivalent to two degrees of freedom spring-mass system model as shown in Fig. 4. With the introduction of sandwich plates, the contact stiffness between the local resonator and the sandwich plate weakens, the resonant frequency of the local resonator decreases, resulting in the decrease of the coupling resonant frequency of the sandwich plate-type elastic metastructures. Owing to the strong coupling of the local resonant Lamb modes of sandwich plates, the resonant mode of the local resonators enhances.

In order to further explain the propagation characteristics and isolation mechanism of the flexural wave in the proposed sandwich plate-type metastructures, the frequency responses of the flexural vibration inside and outside the frequency range of band gap is calculated. Fig. 5(a) shows the propagation characteristics of the flexural wave at 200 Hz, which corresponds to the frequency in the flexural band gap. It can be seen that the flexural wave within the flexural band gap can hardly propagate along the finite array sandwich plate-type metastructures, which proves that the finitely periodic sandwich plate-type metastructures have an excellent weakening effect on the propagation of flexural wave vibration. In contrast, the flexural wave of the structure hardly shows any attenuation at 350 Hz which is beyond the band gap as shown in Fig. 5(b).

3.2. Experimental validation

In an attempt to further verify the correctness of the numerical calculation model, the transmission-measuring experiments of finite array sandwich plate-type metastructures composed of 4 units in the x -direction and 2 units in the y -direction were conducted. The experiment measurement system was composed of an $M + P$ test system, a power amplifier, a computer for installing control and analysis software, three B&K 4507 accelerometers and a vibration exciter, as depicted in Fig. 6 (a). Fig. 6(b) shows the experiment measurement principle.

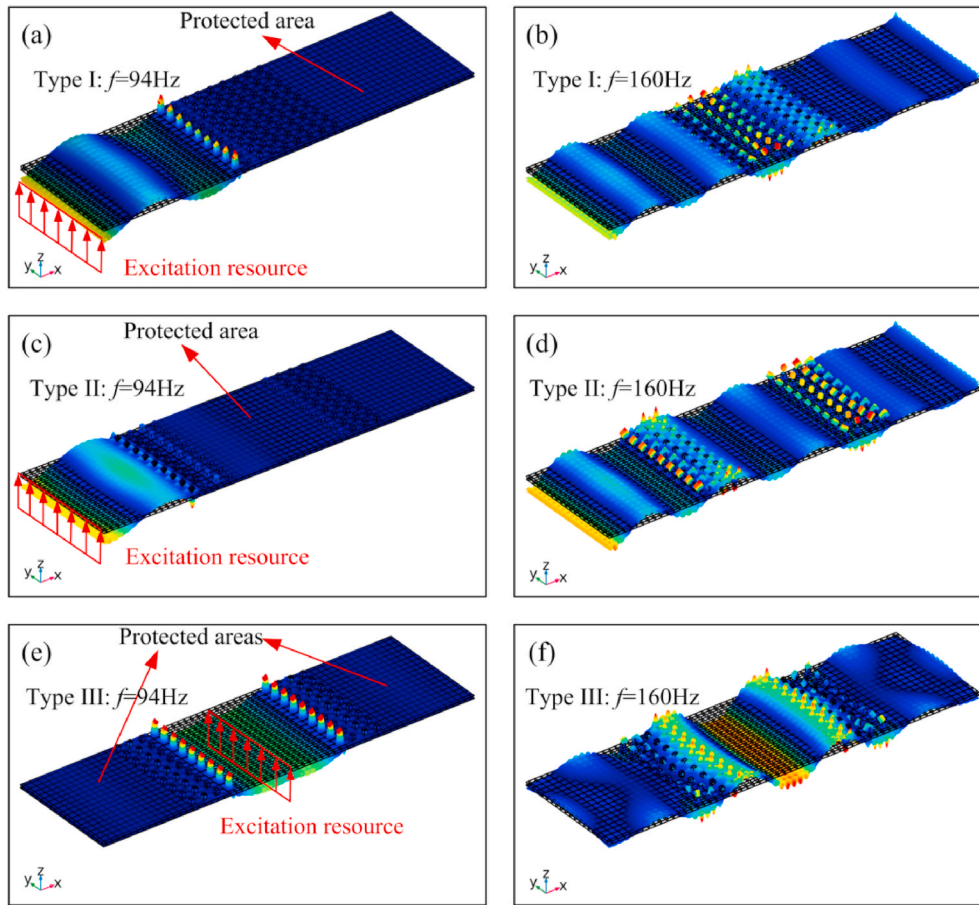


Fig. 11. Displacement fields of three different type applications of full-scale sandwich metastructures.

The experimental sample of the proposed sandwich plate-type elastic metastructure was fabricated as shown in Fig. 6(c). The geometrical dimensions and material parameters of the experimental sample are consistent with those in the numerical simulation. During our experiment test, the flexible rope was used to hang the sandwich plate-type metastructure to achieve free boundary condition. The vibration exciter rod is installed on one side of the metastructure to apply perpendicular excitation to the surface of the sandwich plate. Three accelerometers were installed to measure the input and output acceleration frequency responses. The flexural vibration acceleration frequency response was calculated using the vibration transmission rate formula. The vibration transmission characteristics obtained from the experiment were compared with the numerical calculation results as shown in Fig. 7. It can be seen from Fig. 7 that the flexural vibration frequency response characteristic curve of the sandwich plate-type metastructures had an obvious attenuation area and showed an excellent low frequency flexural vibration isolation performance. The experimental results showed a good agreement with the numerical results, which demonstrated the accuracy and reliability of numerical method. In addition, it is worth noting that the vibration spectra increase fast in some specific frequency range such as 0–10 Hz and 110–150 Hz. This phenomenon can be explained as follow: for the calculation of the frequency response of the sandwich plate-type elastic metastructures, sine sweep acceleration excitation in the frequency range from 0 to 500 Hz is defined so that the natural vibration modes will be stimulated in all frequency range. However, in the actual application, the sandwich plate-type elastic metastructures are generally designed to suppress the vibration propagation in the frequency range of certain band according to the actual excitation resource. According to the forced vibration theory, the vibration magnitude away from the

actual excitation resource drastically reduces.

4. Effects of the geometrical parameters on the band gap

In order to investigate the effects of geometrical parameters on the upper and lower frequencies of the flexural wave band gap, a series of numerical calculations of the proposed sandwich plate-type metastructures were performed as shown in Fig. 8. During our calculations, the material parameters of the sandwich plate and the stepped resonator keep unchanged. The effect of the lattice constant of sandwich plate-type metastructures on the band-gap is shown in Fig. 8(a). As can be seen from Fig. 8(a), with the increase of lattice constant, both the start frequency of the lower boundary and the cut-off frequency of the upper boundary move to low frequency, and the absolute width of the band gaps becomes narrower. This phenomenon can be explained that the band gap of elastic metamaterial is mainly attributed to the local resonator of unit cell rather than the structural periodicity of elastic metamaterial. As the increase of the lattice constant, the additional mass and natural frequency of the sandwich plate-type metastructures decline, leading to the moving of the upper and lower edges of the band gap to low frequency range. It can be observed from Fig. 8(b) and (c) that the start and cut-off frequencies diminish with the increase of the height of the web, while increase with the increase of the thickness of the web. As the thickness of the web increases and the height decreases, the overall spring stiffness of the sandwich plate increases, which results in the increase of the resonant eigenmodes of local resonators. Similarly, we can see from Fig. 8(d) that the increase of plate thickness will also bring about the increase of start and cut off frequencies and the plate thickness has a more significant influence on the band gap evolution. The above parameter analysis mainly focuses on the structural parameters of the

sandwich plate. The structural parameters of the local resonators will also have a key influence on the natural frequency of the local resonant mode. Therefore, the influence of the geometric parameters of the stepped resonators will be analyzed in the following part. As can be seen from Fig. 8 (e) and (f), with the increase of the height of rubber and stub, both the start frequency and the cut-off frequency move to the low frequency. The increase of the height leads to the reduction of the equivalent stiffness in the local resonance element, causing the band gap position to move down. The above analysis results show that the flexural wave band gaps and vibration attenuation characteristics can be adjusted manually by changing the geometric parameters of the proposed sandwich plate-type elastic metastructures, which can be utilized to realize the low-frequency broadband optimization design of sandwich plate-type elastic metastructures.

5. Vibration control application of marine power system

On the basis of the above research, we carried out the exploration of vibration control real ship application of marine power system. Taking into account the scale ratio, the sandwich plate-type elastic metastructure is converted to the real ship scale. The dimensions of the proposed sandwich plate-type elastic metastructure at the real ship scale are depicted in Table 4.

In order to demonstrate the vibration isolation performance of sandwich plate-type metastructures in the marine power system at the real ship scale, the flexural vibration transmission characteristics are studied by using the Solid-Shell multi-physics coupling method as shown in Fig. 9. It can be observed that the 9×9 local resonators are installed in the middle of the sandwich plate. A sine sweep frequency acceleration excitation source is applied to one side of the sandwich plate to simulate the excitation of the marine power system. The flexural vibration direction of the excitation source is along the Z direction, while the flexural wave propagation direction is along the x-direction.

The flexural vibration frequency responses of sandwich plate-type metastructures at the real ship scale are calculated as shown in Fig. 10. It can be found that there is a significant flexural vibration attenuation zone from 78 Hz to 142 Hz in the frequency response, which indicates that the sandwich plate-type metastructures at the real ship can achieve low frequency ship vibration isolation. The flexural vibration frequency response characteristic curve of test point 1 and test point 2 are basically the same, while the frequency response characteristic curve of sandwich plate-type metastructures with damping is smoother than the metastructures without damping.

On the basis of Type I sandwich plate-type metastructures as shown in Fig. 8, another two types of sandwich plate-type metastructures in ship vibration application are designed. The geometrical parameters of Type II are similar to those of Type I, except that it has a central guard region with 4×9 local resonators on either side of the guard region. Type III has the same structural form as Type II, but the location of the excitation source is different. The excitation source of Type II is on the left, while that of Type III is in the center. In order to more intuitively illustrate the application feasibility of sandwich plate-type metastructures in the real ships, the displacement field of the three types of marine sandwich plate-type metastructures at 94 Hz and 160 Hz are calculated respectively, as shown in Fig. 11. Fig. 11 (a) and (b) are the displacement fields of frequencies in and out of the band gap corresponding to type I, Fig. 11 (c) and (d) are the displacement fields of frequencies in and out of the band gap corresponding to type II, and Fig. 11 (e) and (f) are the displacement fields of frequencies in and out of the band gap corresponding to type III, respectively. Fig. 11 intuitively shows the flow law of flexural vibration wave energy in marine sandwich plate-type metastructures. When the excitation frequency of the excitation source is within the band gap range ($f = 94$ Hz), the full-scale sandwich metastructures can effectively limit the vibration near the excitation source and isolate the elastic wave propagation from the excitation source to the protected area. When the excitation frequency of

the excitation source is outside the band gap range ($f = 160$ Hz), the elastic wave can propagate smoothly from the excitation source to the protected area. According to Figs. 10 and 11, it can be found that the full-scale marine sandwich plate-type metastructures have a flexural wave band gap and vibration suppression zone in the frequency range of 78–142 Hz, which can be effectively used to isolate vibration propagation and reduce the vibration and noise level of the protected area. Consequently, it can be concluded that the full scale marine sandwich plate-type metastructures may play an important role in low frequency ship vibration and noise reduction engineering applications.

6. Conclusions

To solve the low frequency vibration and noise control of marine power system, a ship vibration isolation method using the band gap concept of sandwich plate-type elastic metastructures is proposed. An efficient finite element method combination with Solid-Shell multi-physics coupling model and Bloch theory is applied to study the dispersion relations and flexural vibration isolation characteristics of sandwich plate-type elastic metamaterials. The experimental validation of flexural vibration isolation performance was carried out. In addition, the formation mechanism and geometrical parameter study of flexural wave band gap are performed. Finally, the potential application of the proposed ship vibration isolation method in the marine power system is explored. The following conclusions can be drawn:

- (1) The proposed sandwich plate-type elastic metastructures possess significant low-frequency subwavelength flexural wave band-gap and vibration isolation characteristics under the premise of structural strength and stiffness. The experimental results are consistent with the numerical results, which demonstrate the accuracy and reliability of numerical method.
- (2) The formation mechanism of flexural band gap is mainly attributed to the mode coupling of the local resonant mode and the Lamb wave mode of sandwich plate. The subwavelength flexural wave band-gap can be effectively modulated by the geometrical parameters.
- (3) Full-scale marine sandwich plate-type metastructures have a flexural wave band gap and vibration suppression zone in the frequency range of 78–142 Hz, which demonstrates that the proposed ship vibration isolation method has an attractive application prospect in low frequency vibration and noise control of marine power system.

CRediT authorship contribution statement

Dingkang Chen: Formal analysis, Writing – original draft. **Huan Zi:** Formal analysis, Writing – review & editing. **Yinggang Li:** Supervision, Funding acquisition, Writing – review & editing. **Xunyu Li:** Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Badreddine Assouar, M., Oudich, M., 2012. Enlargement of a locally resonant sonic band gap by using double-sides stubbed phononic plates. *Appl. Phys. Lett.* 100 (12), 123506.
- Bilal, O.R., Hussein, M.I., 2013. Trampoline metamaterial: local resonance enhancement by springboards. *Appl. Phys. Lett.* 103 (11), 111901.
- Cai, Y., Hui Wu, J., Xu, Y., Ma, F., 2021. Realizing polarization band gaps and fluid-like elasticity by thin-plate elastic metamaterials. *Compos. Struct.* 262, 113351.
- Fang, N., Xi, D., Xu, J., Ambati, M., Srituravanich, W., Sun, C., Zhang, X., 2006. Ultrasonic metamaterials with negative modulus. *Nat. Mater.* 5 (6), 452–456.
- Jung, J., Goo, S., Wang, S., 2020. Investigation of flexural wave band gaps in a locally resonant metamaterial with plate-like resonators. *Wave Motion* 93, 102492.
- Kandasamy, R., Cui, F., Townsend, N., Foo, C.C., Guo, J., Shenoi, A., Xiong, Y., 2016. A review of vibration control methods for marine offshore structures. *Ocean. Eng.* 127, 279–297.
- Langfeldt, F., Gleine, W., 2019. Membrane- and plate-type Acoustic metamaterials with elastic unit cell edges. *J. Sound Vib.* 453, 65–86.
- Lee, S.H., Park, C.M., Seo, Y.M., Wang, Z.G., Kim, C.K., 2010. Composite acoustic medium with simultaneously negative density and modulus. *Phys. Rev. Lett.* 104 (5), 54301.
- Li, Y., Chen, T., Wang, X., Xi, Y., Liang, Q., 2015. Enlargement of locally resonant sonic band gap by using composite plate-type Acoustic metamaterial. *Phys. Lett.* 379 (5), 412–416.
- Li, Y., Zhou, Q., Zhou, L., Zhu, L., Guo, K., 2019. Flexural wave band gaps and vibration attenuation characteristics in periodic Bi-directionally orthogonal stiffened plates. *Ocean. Eng.* 178, 95–103.
- Li, Y., Zhu, L., Chen, T., 2017. plate-type elastic metamaterials for low-frequency broadband elastic wave attenuation. *Ultrasonics* 73, 34–42.
- Lin, T.R., Pan, J., O'Shea, P.J., Mechefske, C.K., 2009. A study of vibration and vibration control of ship structures. *Mar. Struct.* 22 (4), 730–743.
- Liu, Z., Chan, C.T., Sheng, P., 2005. Analytic model of phononic crystals with local resonances. *Phys. Rev. B Condens. Matter* 71 (1), 14101–14103.
- Liu, Z., Zhang, X., Mao, Y., Zhu, Y.Y., Yang, Z., Chan, C.T., Sheng, P., 2000. Locally resonant sonic materials. *Science* 289 (5485), 1734–1736.
- Miranda Jr., E.J.P., Nobrega, E.D., Rodrigues, S.F., Aranas Jr., C., Dos Santos, J.M.C., 2020. Wave attenuation in elastic metamaterial thick plates: analytical, numerical and experimental investigations. *Int. J. Solid Struct.* 204–205, 138–152.
- Peng, H., Frank Pai, P., 2014. Acoustic metamaterial plates for elastic wave absorption and structural vibration suppression. *Int. J. Mech. Sci.* 89, 350–361.
- Pennec, Y., Djafari-Rouhani, B., Larabi, H., Vasseur, J.O., Hladky-Hennion, A.C., 2008. Low-frequency gaps in a phononic crystal constituted of cylindrical dots deposited on a thin homogeneous plate. *Phys. Rev. B* 78 (10).
- Ruan, Y., Liang, X., Hua, X., Zhang, C., Xia, H., Li, C., 2021. Isolating low-frequency vibration from power systems on a ship using spiral phononic crystals. *Ocean. Eng.* 225, 108804.
- Sheng, P., Mei, J., Liu, Z., Wen, W., 2007. Dynamic mass density and acoustic metamaterials. *Phys. B Condens. Matter* 394 (2), 256–261.
- Song, Y., Feng, L., Liu, Z., Wen, J., Yu, D., 2019. Suppression of the vibration and sound radiation of a sandwich plate via periodic design. *Int. J. Mech. Sci.* 150, 744–754.
- Wang, Q., Li, J., Zhang, Y., Xue, Y., Li, F., 2021. Bandgap properties in metamaterial sandwich plate with periodically embedded plate-type resonators. *Mech. Syst. Signal Process.* 151, 107375.
- Wu, T., Huang, Z., Tsai, T., Wu, T., 2008. Evidence of complete band gap and resonances in a plate with periodic stubbed surface. *Appl. Phys. Lett.* 93 (11), 111902.
- Wu, Z., Li, F., Zhang, C., 2015. Vibration band-gap properties of three-dimensional kagome lattices using the spectral element method. *J. Sound Vib.* 341, 162–173.
- Xiao, Y., Wen, J., Wen, X., 2012. Flexural wave band gaps in locally resonant thin plates with periodically attached spring-mass resonators. *J. Phys. Appl. Phys.* 45 (19), 195401.
- Yuan, X., Chen, M., Yao, Y., Guo, X., Huang, Y., Peng, Z., Xu, B., Lv, B., Tao, R., Duan, S., Liao, H., Yao, K., Li, Y., Lei, H., Chen, X., Hong, G., Fang, D., 2021. Recent progress in the design and fabrication of multifunctional structures based on metamaterials. *Curr. Opin. Solid State Mater. Sci.* 25 (1), 100883.
- Zhang, B., Han, Q., Zhang, X., 2017. Recent advances in vibration control of offshore platforms. *Nonlinear Dynam.* 89 (2), 755–771.